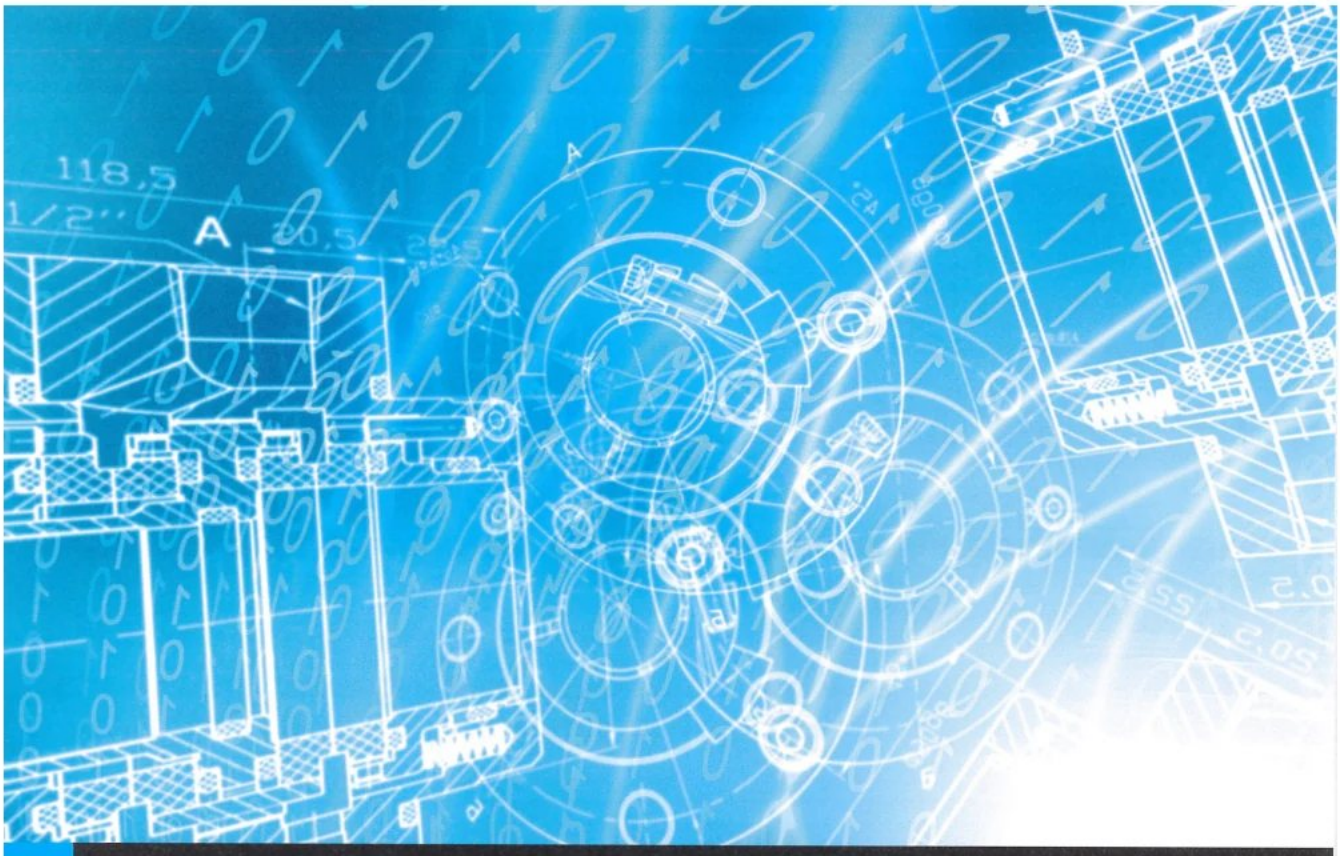


Chapter 7:
Monitoring and control systems

Learning objectives

By the end of this chapter you should be able to:

- show an understanding of monitoring and control systems
- show understanding of how bit manipulation can be used to monitor/control a device.



7.01 Monitoring systems

A monitoring system can be used to create a record of the condition of a system over a period of time. A monitoring system is used more often to detect when a particular physical property of a system goes outside a desired range; for example, if the CPU is too hot.

Discussion Point:

Set yourself a time limit of one minute. During this minute, by considering what measurement will be involved, ask yourself how many different types of monitoring system you can identify.

Let's consider temperature as an example. If this was being monitored under human control, the measurement could be made with a standard mercury thermometer. However, in this chapter we are interested in systems where a computer or microprocessor is being used. These systems require a measuring device that records a value which can be transmitted to the computer. Such a measuring device is a called a **sensor**. An example of a sensor for measuring temperature is a thermocouple, which outputs an electrical voltage that changes with temperature.

It is important to understand that in a monitoring system, a sensor does not have any built-in intelligence, so it cannot take any action if there is a problem. If the temperature measured becomes dangerously high it is the computer that sounds an alarm.

There are a wide variety of sensors available. For some the name indicates the property being measured such as pressure, humidity, carbon monoxide, pH or sound. For others such as an infrared sensor there are different methods of use. A passive infrared sensor just measures the level of infrared light received. In other cases, there is transmission of infrared light with the sensor possibly measuring the level of the light that is reflected back. Other sensors are given a generic name such as a motion sensor, for which different examples will be measuring different physical properties.

Question 7.01

How many different types of motion sensor are you aware of?

7.02 Control systems

A control system has the monitoring activity plus the capability to control a system. The control element of a monitoring and control system needs a device called an **actuator**. An actuator is an electric motor that is connected to a controlling device. It might be used for switching on or off or for adjusting a setting.



KEY TERMS

Sensor: a hardware device that measures a property and transmits a value to a controlling computer

Actuator: a hardware device that receives a signal from a computer and adjusts the setting of a controlling device

Discussion Point:

Refer back to the examples you identified as monitoring systems. How many were actually control systems? If they were monitoring systems could they be modified to become control systems?

Figure 7.01 shows a schematic diagram of a computer-controlled environment.

Note that Figure 7.01 includes an analogue-to-digital converter (ADC) and a digital-to-analogue converter (DAC) as separate components. In a real system they are likely to be integral to the sensor or actuator device.

For the system shown in Figure 7.01 there is a continuing process where the computer at regularly timed intervals signals the sensor to provide a measurement. If the measurement value received by the computer is not in the desired range the computer initiates a control action. The next timed measurement will happen after this control action has taken place. In effect this next measurement provides feedback to the computer on the effect of the control action. Feedback is essential in a control system.

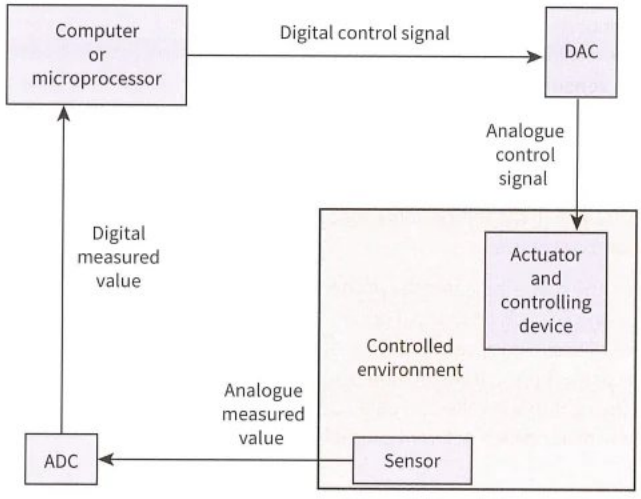


Figure 7.01 Computer-controlled environment

TIP

You need to remember that a sensor does not have any built-in intelligence so it cannot itself take any action if a problem occurs.

A closed-loop feedback control system is a special type of monitoring and control system where the feedback directly controls the operation. Figure 7.02 shows a schematic diagram of such a system. A microprocessor functions as the controller. This compares the value for the actual output, as read by the sensor, with the desired output. It then transmits a value to the actuator which depends on the difference calculated.

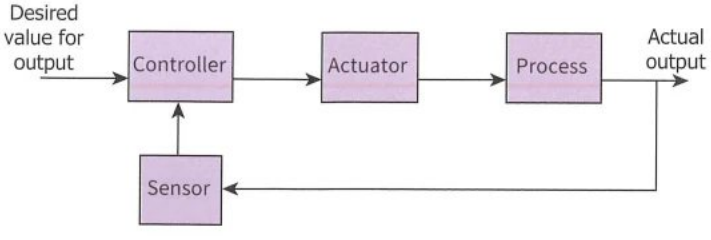


Figure 7.02 Closed-loop feedback control system